

BONUS MISSION – MALICIOUS WAYPOINT

Base Mission

The 7-waypoint mission from Mission 1 was used as the base mission — launching from Tech Green and visiting Bobby Dodd Stadium, Couch Park, CRC, Van Leer, Coliseum, and Russ Chandler, before returning to launch. This full mission was executed normally as a baseline, confirming standard autonomous operation before the attack was introduced. The complete normal mission execution is documented in **Task_6_Base_Mission.mp4**.

Attack Scenario

Attack Target: WP4 (Van Leer)

The attack execution, including the drone abandoning its mission at WP4, navigating to the injected coordinate 200m south, and completing full touchdown and disarm, is documented in **Task_6_attack.mp4**.

Rather than modifying mission items directly, the attack was implemented via **GUIDED mode command injection** using a Python script (`attack.py`). When the drone reached WP4 (Van Leer), the script issued the following commands over MAVLink:

1. Switched the drone into GUIDED mode, interrupting the autonomous mission mid-flight
2. Issued a `simple_goto` command redirecting the drone to a position 200m south of Van Leer
3. Issued a `LAND` command upon reaching the injected coordinate, forcing an unplanned landing

Drone response:

- Completed WP1 (Bobby Dodd), WP2 (Couch Park), WP3 (CRC) normally
- Abandoned the original mission upon reaching WP4 (Van Leer)
- Navigated 200m south to the injected coordinate
- Initiated and completed landing at the unauthorized location, never reaching WP5 (Coliseum), WP6 (Russ Chandler), or RTL

Threat Model

Attacker Capabilities:

- Access to GCS or MAVLink communication channel
- Ability to inject MAVLink GUIDED mode commands and `simple_goto` / `LAND` messages
- Knowledge of mission structure and waypoint timing

Attack Type: Insider threat / Compromised GCS / MAVLink channel interception

Impact:

- Mission manipulation via command injection at WP4, leaving WP5, WP6, and RTL never executed
- Forced unplanned landing at attacker-controlled location 200m south of Van Leer
- Complete loss of autonomous mission control from WP4 onward

CIA Defense Strategy

Confidentiality

- Encrypt MAVLink communication channels to prevent interception and eavesdropping
- Use secure telemetry modules to prevent unauthorized channel access

Integrity

- **MAVLink2 message signing** , cryptographically signs each MAVLink packet, preventing injection of unauthorized commands such as the `simple_goto` and `LAND` used in this attack
- **Command authorization and whitelisting** , restrict which command types (e.g., mode changes, `LAND`) are accepted from which sources, blocking unauthenticated GUIDED mode switches mid-mission
- Digital signing of mission files to detect tampering
- Mission checksum validation to verify no waypoints were altered in transit

Availability

- Redundant control links to maintain legitimate GCS authority even under attack
- Failsafe override logic to detect unexpected mode changes mid-mission
- Parameter lock to prevent unauthorized flight mode transitions

Conclusion

The attack demonstrated how MAVLink command injection, specifically GUIDED mode takeover via `simple_goto` and `LAND` can override autonomous operations mid-mission without ever modifying the mission file directly. In this case, a 7-waypoint mission was successfully hijacked at WP4 (Van Leer), causing the drone to abandon the remaining waypoints and land at an unauthorized location. This is a realistic and stealthy attack vector, as it exploits the live command channel rather than stored mission data.

Implementing CIA principles, particularly MAVLink2 signing and command whitelisting, directly counters this class of injection attack by ensuring only authenticated, authorized commands are accepted by the flight controller.